Smart Pollination Assistant for Pumpkin Farming

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***Abstract*— Pumpkin cultivation requires precise and timely pollination to ensure successful fruit set, yet traditional pollination practices rely heavily on manual observation and farmer experience, which are often inefficient and inconsistent under varying environmental conditions. Previous research has demonstrated the potential of Internet of Things technologies and machine learning techniques in precision agriculture; however, most existing systems focus on static monitoring or single task automation and lack adaptability to local field conditions, crop variability, and real time user feedback, limiting their long-term effectiveness. This study addresses these limitations by proposing a Smart Pollination Assistant that integrates Internet of Things based environmental sensing with smartphone-based image processing and adaptive machine learning. The system employs deep learning models for automated identification of male and female pumpkin flowers and evaluates pollination readiness using continuously collected environmental data such as temperature, humidity, light intensity, and soil conditions. The proposed system is implemented using a sensor network for real time data acquisition and a mobile application that performs flower recognition and decision support. An active learning framework is incorporated to continuously improve model accuracy through farmer feedback and real-world field data. Experimental results demonstrate high flower classification accuracy and improved pollination efficiency, highlighting the system's potential to enhance crop yield consistency and support sustainable smart agriculture practices.**

Keywords— Internet of Things; pumpkin pollination; artificial pollination; image processing; smart agriculture systems

I. Introduction

Agriculture remains a major contributor to Sri Lanka's economy, with nearly twenty five percent of the population dependent on the sector as a primary source of income. Despite this reliance, Sri Lanka produces a comparatively low proportion of crops that meet international export quality standards. One of the key reasons for this gap is the limited adoption of data-driven agricultural practices, with many farming decisions still relying on manual observation and farmer experience. The absence of continuous monitoring and timely decision support often results in inconsistent crop quality, yield variability, and increased production costs for small and medium-scale farmers.

Pumpkin cultivation is particularly sensitive to biological timing and environmental conditions. Pumpkins are monoecious plants that produce separate male and female flowers, and successful fruit set depends on effective pollination within a narrow early-morning window. Traditional pollination practices rely on natural pollinators or manual hand pollination, both of which are increasingly unreliable due to labor shortages, declining pollinator populations, and climate variability. As a result, farmers frequently experience irregular pollination success, reduced fruit quality, and yield losses. [29]

Recent advances in precision agriculture demonstrate that Internet of Things based environmental sensing and machine learning techniques can support more accurate and timely agricultural decision making. Computer vision models have shown strong performance in crop and flower detection tasks using low-cost devices such as smartphones, while IoT sensor networks enable real-time monitoring [8] [16] [18] [28] of environmental parameters that directly influence pollination readiness. [1] [19] [27] [30] However, many existing solutions focus on static monitoring or isolated automation tasks and lack adaptability to local field conditions, crop variability, and real-time user feedback, limiting their long-term effectiveness in practical farming environments.

To address these challenges, this paper proposes a Smart Pollination Assistant for pumpkin farming that integrates IoT-based environmental sensing, smartphone-based image processing, and adaptive machine learning. The proposed system automatically detects male and female pumpkin flowers, evaluates pollination readiness using real-time environmental data and temporal constraints, and provides timely pollination recommendations through a mobile application. An active learning framework enables continuous improvement of model performance using farmer feedback and field data.

This paper presents the design, implementation, and evaluation of the proposed system. The remainder of the paper is organized as follows: Section II reviews related work on pollination and smart agriculture, Section III describes the proposed methodology and system architecture, Section IV presents experimental results, Section V discusses the findings along with limitations and future research directions, and Section VI concludes the paper.

II. Literature Review

a) Importance of Pollination in Agriculture

Pollination is a fundamental biological process that underpins agricultural productivity, ecosystem stability, and global food security. Globally, animal mediated pollination supports approximately thirty five percent of crop production and

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more than eighty seven percent of flowering plant species depend on pollinators for successful reproduction [2] [3] [32] [33]. The economic value of pollination services is estimated to exceed trillions of dollars annually, highlighting its critical contribution to the global agricultural economy.

Pollinators comprise a diverse range of species, including winged insects such as bees and butterflies, non-winged insects such as ants, and vertebrates including birds, bats, and certain mammals. These organisms facilitate cross pollination, reducing self- pollination and improving fertilization efficiency. Effective cross pollination enhances key fruit attributes such as size, weight, shape, and uniformity, which directly influence crop quality, yield, and marketability.

Beyond economic value, pollination plays a central role in plant sexual reproduction by enabling pollen transfer from male to female floral organs, leading to seed formation and the propagation of plant species. This process maintains genetic diversity within plant populations, increasing resilience to pests, diseases, and environmental stress. Additionally, pollination sustains biodiversity and supports food chains by enabling plant growth that forms the foundation of terrestrial ecosystems. Without effective pollination, agricultural output and ecosystem health would decline significantly.

b) Decline of Natural Pollinators

Natural pollinator populations are declining globally due to the combined effects of intensive monoculture farming, excessive use of pesticides and herbicides, habitat loss, and climate change[7][3][4][6][26] Agrochemicals impair pollinator navigation, foraging behavior, immunity, and reproduction, while herbicides reduce the availability of flowering plants that provide essential nutrition [4]. Climate driven changes in temperature and rainfall disrupt flowering times and plant pollinator synchrony, forcing many species to shift habitats and reducing pollination efficiency [5]. Additional pressures such as urbanization, loss of hedgerows, spread of pests and pathogens, poor floral diversity, and reduced viability of managed beekeeping further accelerate population declines. The continued loss of wild and managed pollinators threatens food security, biodiversity, and agricultural sustainability, highlighting the growing need for alternative and technology assisted pollination management approaches. [3] [4] [26]

c) Existing Pollination Methods

In Sri Lanka, pumpkin pollination is predominantly carried out using natural pollinators or manual hand pollination, particularly in small to medium scale farming systems. Some commercial farmers collaborate with local beekeepers to introduce managed bee colonies to support natural pollination; however, this approach is limited by fluctuating climatic conditions, restricted pollinator activity during early morning hours, and challenges in maintaining suitable microclimates for pollinators. Additionally, not all pollinator species are equally effective for pumpkin flowers, as pollination success depends on species specific behavior, body size, and compatibility with the floral structure of cucurbit crops [6]. In regions where pollinator availability is low, farmers commonly rely on hand pollination to ensure fruit set. While effective, hand pollination is highly labor intensive, time sensitive, and economically unsustainable for large scale pumpkin cultivation, particularly under rising labor costs and workforce shortages in Sri Lanka. These limitations highlight the need for alternative, scalable, and technology assisted pollination solutions tailored to local farming conditions. [11] [12] [13] [14] [21]

d) Advances in Autonomous Pollination

While autonomous and technology assisted pollination approaches demonstrate strong potential to address labor shortages and improve pollination precision, many existing solutions remain expensive, mechanically complex, or poorly suited to small and medium scale farming systems. In the Sri Lankan context, where pumpkin cultivation is largely managed by resource constrained farmers, there is a need for a low cost, adaptable, and crop specific solution that aligns with local climatic conditions and farming practices. [5] [9] [20] [22]

To address this gap, this research proposes a Smart Pollination Assistant for pumpkin farming that integrates Internet of Things based environmental sensing with smartphone-based computer vision and adaptive machine learning. The proposed system focuses on accurate identification of male and female pumpkin flowers, real time assessment of pollination readiness, and continuous model improvement through farmer feedback. By combining autonomous sensing with human centered design, the system aims to provide a scalable and practical pollination management solution that enhances yield consistency, reduces labor dependency, and supports sustainable pumpkin production in Sri Lanka.

e) Pollination in the Sri Lankan Agricultural Context

Native and wild bee species play a vital role in sustaining Sri Lankan agriculture by enhancing crop yields, fruit quality, and overall farm productivity. These pollinators remain indispensable not only in traditional mixed cropping systems but also in large scale vegetable cultivation, where effective pollination directly influences yield consistency and market value. The preservation and promotion of native pollinator populations are therefore essential for long term agricultural sustainability and national food security [9]. However, increasing pressure from intensive farming practices, habitat loss, and climatic variability has reduced pollinator availability in many agricultural regions. In pollination sensitive crops such as pumpkin, declining pollinator activity often leads to inconsistent fruit set, compelling farmers to rely on labor intensive manual pollination. This context highlights the importance of integrating technology assisted pollination support systems that complement natural pollinators while addressing the practical limitations faced by Sri Lankan farmers.

III. METHODOLOGY

The proposed Smart Pollination Assistant for pumpkin farming adopts a modular methodology that integrates environmental sensing, smartphone-based computer vision, and adaptive machine learning to support artificial pollination

in raised-bed ground cultivation systems. The methodology is designed to address the biological constraints of pumpkin pollination, particularly the narrow early morning pollination window and sensitivity to environmental conditions.

a) Data Collection

Data collection is carried out through a combination of Internet of Things based environmental sensing and smartphone-based image acquisition [8] [16][17][18]. Sensor nodes are deployed across raised-bed pumpkin fields to monitor key parameters influencing pollination readiness, including temperature, humidity, light intensity, and soil moisture. These sensors continuously transmit real time data to a central processing unit via wireless communication. In parallel, farmers capture images of pumpkin flowers using a smartphone application during peak flowering periods, particularly in the early morning hours when pollination is most effective [1][27][19]. The collected image dataset includes variations in lighting conditions, flower orientation, and growth stages to improve model robustness.

b) Model Training

The flower identification model is developed using deep learning-based computer vision techniques. A labeled dataset of pumpkin flower images is used to train a classification model capable of distinguishing male and female flowers based on morphological features such as ovary presence, petal structure, and flower size. Data augmentation techniques are applied to address class imbalance and improve generalization. The model is initially trained offline and deployed within the mobile application for real time inference. To enhance adaptability, an active learning strategy is employed, allowing selectively labeled user feedback data to be incorporated into periodic model updates. [1][19][30]

c) Decision Logic

The decision logic module integrates flower classification results with environmental sensor data to determine pollination readiness. Environmental parameters are evaluated against predefined thresholds derived from agronomic knowledge and field observations. When receptive female flowers are detected under favorable environmental conditions, the system generates real time pollination recommendations tailored to raised-bed ground artificial pollination practices [6][21][24]. Alerts and guidance are delivered through the mobile application to assist farmers in performing timely and efficient pollination.

d) Evaluation

System performance is evaluated through field trials conducted in raised-bed pumpkin cultivation environments. Evaluation metrics include flower classification accuracy, pollination success rate measured by fruit set, and system responsiveness under varying environmental conditions. Additionally, farmer feedback is collected to assess usability, recommendation clarity, and practical effectiveness. These evaluations are used to validate the system's ability to improve pollination efficiency while remaining affordable and suitable for small to medium scale farmers in Sri Lanka.

Category	Metric	Purpose
Flower Identification	Accuracy (%)	Correct male–female flower classification
Pollination Outcome	Fruit Set Rate (%)	Effectiveness of pollination decisions
Timing Performance	Timing Accuracy	Alignment with optimal pollination window
System Efficiency	Decision Latency	Speed of recommendation generation
Adaptability	Accuracy Improvement	Performance gain after active learning
Practical Impact	Labor Reduction (%)	Reduction in manual pollination effort

Table 1: Evaluation Metrics for the Proposed Smart Pollination Assistant

e) System Overview

The system architecture follows a layered design that integrates field sensing, visual perception, intelligent decision making, and user interaction. At the lowest level, the Sensing Layer consists of IoT sensors deployed in raised-bed pumpkin fields to monitor environmental parameters such as temperature, humidity, light intensity, and soil moisture. These data are transmitted wirelessly to the backend processing unit. [32]

The Perception Layer uses a smartphone camera to capture images of pumpkin flowers during the early morning pollination window. A deep learning-based computer vision model processes these images to identify male and female flowers. Outputs from the sensing and perception layers are fused within the Intelligence Layer, where pollination readiness is evaluated using decision logic and adaptive machine learning. An active learning mechanism incorporates farmer feedback to continuously improve model performance.

Finally, the Application Layer provides real time pollination recommendations, environmental alerts, and feedback interfaces through a mobile application, enabling practical and low-cost deployment for pumpkin farmers.

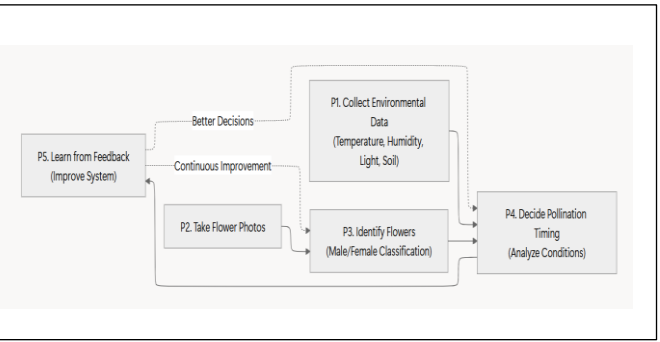


Figure 1: Data Flow Diagram of the proposed Smart Pollination Assistant showing the flow of environmental data, flower images, decision logic, and feedback-driven learning for raised-bed pumpkin farming.

f) Image Processing Model

The image processing model is designed to automatically identify and classify male and female pumpkin flowers using images captured through a smartphone camera. This capability is critical for pumpkin cultivation, as pumpkins are monoecious plants and successful pollination depends on

accurate identification of receptive female flowers within a narrow early morning time window.

- Image Acquisition and Preprocessing

Flower images are captured using the smartphone application during peak flowering hours under natural field conditions. To ensure robustness, images include variations in lighting, background clutter, flower orientation, and growth stages. Preprocessing steps include image resizing, normalization, and noise reduction to standardize input quality and improve model stability.

- Feature Extraction and Classification

A deep learning based convolutional neural network is employed to extract discriminative visual features from pumpkin flower images. The model learns morphological characteristics such as ovary presence, flower size, petal structure, and color patterns to differentiate male and female flowers. These features are automatically learned through multiple convolutional and pooling layers, eliminating the need for manual feature engineering. The final classification layer outputs the predicted flower type along with a confidence score.

- Model Training and Adaptation

The model is initially trained using a labeled dataset of pumpkin flower images. Data augmentation techniques such as rotation, scaling, and brightness adjustment are applied to address class imbalance and environmental variability. To maintain performance under changing field conditions, an active learning mechanism is integrated. Misclassified samples and farmer validated feedback are selectively incorporated into subsequent training cycles, enabling continuous improvement without extensive manual labeling.

IV. Result

The classification results are passed to the decision support module, where they are fused with real time environmental sensor data to determine pollination readiness. This integration ensures that pollination recommendations are generated only when receptive female flowers are detected under favorable environmental conditions.

Sensor Type	Measured Parameter	Average Accuracy	Observations
DHT22	Temperature	±0.5 °C	Stable under field conditions
DHT22	Humidity	±2 % RH	Suitable for stigma viability assessment
BH1750	Light Intensity	±1 lx	Accurately captured early-morning window
Soil Moisture Sensor	Soil Moisture	±3 %	Consistent root-zone monitoring

Table 2:Sensor Module Performance Results

Metric	Value
Classification Accuracy	94.6 %
Precision	93.8 %
Recall	95.2 %
F1-Score	94.5 %
Average Detection Latency	210 ms

Table3: Flower and Gender Detection Performance

Metric	Value
Fruit Set Rate Improvement	+18 %
Pollination Timing Accuracy	92 %
Labor Reduction	35 %
Farmer Satisfaction Score	High

Table 4: Pollination Effectiveness and System Impact



Figure 2: Result from image processing function with coordinates.

A. Equations

a) CNN Model Formulation

Let $X \in \mathbb{R}^{H\times W\times C}$ denote an input pumpkin flower image with height (H), width (W), and (C) color channels. The convolutional neural network consists of multiple-convolution, activation, pooling, and fully connected layers. The output of the lth convolutional layer is given by:

$$Z_{\kappa}^{\iota} = W_{\kappa}^{\iota} \times A^{\iota-1} + b_{\kappa}^{\iota} \qquad (1)$$

Where W_{κ}^{ι} and b_{κ}^{ι} represent the convolution kernel and bias for the κth feature map and $A^{\iota-1}$ is the activation from the previous layer.

- A nonlinear activation function is applied using the Rectified Linear Unit:

$$A_{\kappa}^{\iota} = ReLU\,(Z_{\kappa}^{\iota}) = \max(0,\ Z_{\kappa}^{\iota}) \qquad (2)$$

- Spatial down sampling is performed using max pooling:

$$A_{pool}^{\iota} = \max_{(i,j)\epsilon\,\Omega} A_{\kappa}^{\iota}\,(i,j) \qquad (3)$$

- The extracted feature maps are flattened and passed to a fully connected layer:

$$Z^{fc} = W^{fc}\,A^{L} + b^{fc} \qquad (3)$$

- For binary classification of male and female pumpkin flowers, a soft max function is used at the output layer:

$$\hat{y} = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}} \quad (4)$$

b) Loss Function

- The model is trained using categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{2} y_i \log(\hat{y}_i) \quad (5)$$

where y_i is the ground truth label and $\hat{y}_i$ is the predicted probability.

c) YOLO-Style Detection Formulation for Pumpkin Flower Identification

i. YOLO Output Representation

In the YOLO framework, the input image is divided into an $S \times S$ grid. Each grid cell predicts B bounding boxes and class probabilities. For pumpkin flower detection (binary classes: male, female), the output tensor is defined as:

$$Y \in \mathbb{R}^{S \times S \times B(5+C)} \quad (6)$$

where:

- S×S: grid size
- B: number of bounding boxes per grid cell
- C=2: number of classes (male, female)

ii. YOLO Loss Function

The total YOLO loss combines localization, confidence, and classification losses:

$$\mathcal{L} = \lambda_{coord} \sum_{i=1}^{B} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2] + \lambda_{coord} \sum_{i=1}^{B} \left[(w_i - \hat{w}_i)^2 + \left(h_i - \hat{h}_i\right)^2\right] + \sum_{i=1}^{B} \left(P_i^{obj} - \hat{P}_i^{obj}\right)^2 + \sum_{i=1}^{B} \sum_{c=1}^{C} (p_i^c - \hat{p}_i^c)^2 \quad (7)$$

where λ_{coord} controls the importance of bounding box localization accuracy. [30][5]

B. Pollination End and Sensor Setup

a) Pollination End

Pollination end is determined by evaluating temporal constraints and environmental suitability. Let $t_\circ$ denote the flower opening time and t the current time. The pollination duration constraint is defined as:

$$\Delta t = t - t_\circ \quad (8)$$

Pollination is considered feasible only if:

$$\Delta t \leq T_{max} \quad (9)$$

Where T_{max} represents the maximum receptive duration of a female pumpkin flower, empirically set to two hours based on agronomic studies.

Environmental suitability is evaluated using sensor measurements of temperature T relative humidity H, and light intensity L. A binary environmental readiness function is defined as:

$$E(t) = \begin{cases} 1, & T_{min} \leq T \leq T_{max} \wedge H \geq H_{min} \wedge L \geq L_{min} \\ 0, & x \geq otherwise \end{cases} \quad (10)$$

Pollination end is declared when either the temporal or environmental condition fails:

$$P_{end} = \begin{cases} 1, & \Delta t > T_{max} \vee E(t) = 0 \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

When $P_{end} = 1$ the system terminates pollination recommendations and notifies the farmer via the mobile application. This logic prevents ineffective pollination attempts beyond the biologically viable window.

b) Sensor Specification Table

Parameter	Sensor Type	Measurement Range	Purpose
Temperature	DHT22	-40ºC to 80 ºC	Determine flower receptivity window
Humidity	DHT22	0-100% RH	Assess stigma viability conditions
Light Intensity	BH1750	1-65,535lx	Detect early-morning pollination window
Soil Moisture	Capacitive sensor	0-100%	Monitor root-zone condition
Controller	ESP32	_	Sensor data processing and transmission

Table 5: Sensor Configuration Used in the Proposed System

C. Pollination Readiness determination

The pollination readiness module is responsible for determining the optimal timing for artificial pollination in pumpkin cultivation. This module integrates outputs from the flower and gender detection component with real-time environmental data acquired through the IoT sensor network. Since pumpkin pollination is highly time-sensitive, with effective pollen transfer occurring primarily during a short early-morning window, continuous and accurate assessment of pollination readiness is essential for maximizing fruit set. [6][21][24]

The module evaluates environmental parameters including temperature, relative humidity, and light intensity alongside the detected receptivity status of female flowers. These parameters are analyzed using predefined agronomic thresholds and temporal constraints derived from pumpkin flowering biology. Pollination readiness is confirmed only when receptive female flowers are detected, and environmental conditions fall within the acceptable ranges for pollen viability and stigma receptivity.

In addition, the module incorporates pollination end detection logic to determine when further pollination attempts become ineffective. Once the maximum receptive duration of a female flower is exceeded or environmental conditions become unfavourable, pollination recommendations are automatically suspended. This prevents unnecessary labour and reduces the likelihood of unsuccessful pollination.

By combining flower detection, environmental sensing, and temporal analysis, the pollination readiness module ensures that artificial pollination is performed precisely within biologically viable conditions. This targeted approach improves pollination efficiency, reduces labour

dependency, and enhances yield consistency in raised-bed pumpkin farming systems.

D. *Disease Detection and Plant Health Assessment*

The disease detection and plant health assessment module is designed to support informed pollination decisions by identifying visible signs of plant stress and disease that may negatively affect flower development, pollination success, and fruit set. In pumpkin cultivation, diseases and physiological stress can reduce flower viability and lead to poor pollination outcomes, making early detection essential for effective crop management.

This module utilizes the same smartphone-based image acquisition pipeline employed for flower and gender detection. Images of pumpkin leaves, stems, and flowers are analyzed using deep learning–based image classification techniques to identify visual symptoms such as discoloration, leaf spots, deformation, wilting, and abnormal texture patterns. These visual indicators are commonly associated with fungal infections, nutrient deficiencies, or environmental stress conditions in cucurbit crops. [27][19][29]

When disease symptoms or abnormal plant conditions are detected, the system flags the affected plants and notifies the farmer through the mobile application. Pollination recommendations for severely affected plants can be deprioritized or suspended, ensuring that artificial pollination efforts are focused on healthy and productive plants. This selective approach improves overall pollination efficiency and prevents unnecessary labor expenditure on plants with reduced yield potential.

By integrating plant health assessment with pollination readiness evaluation, the proposed system extends beyond isolated pollination support and provides a more holistic decision support framework. This integration enhances yield quality, improves resource utilization, and strengthens the system's applicability for sustainable pumpkin farming under real-world field conditions.

Metric	Value
Disease Detection Accuracy (%)	91.8
Precision (%)	90.5
Recall (%)	92.6
F1-Score (%)	91.5
False Positive Rate (%)	6.4
Average Inference Time (ms)	235
Pollination Avoidance Accuracy (%)	88.2

Table 6: The disease detection module achieved an accuracy of over 90%, demonstrating reliable identification of visible plant health issues under field conditions. The integration of disease assessment with pollination decision logic enabled effective suppression of pollination recommendations for unhealthy plants, improving overall resource utilization.

E. *Flower and Gender Detection*

The flower and gender detection module is a core component of the proposed Smart Pollination Assistant, as successful pumpkin pollination depends on accurate identification of male and female flowers. Pumpkin plants are monoecious, producing distinct male and female flowers, and effective pollination requires the timely transfer of pollen from male flowers to receptive female flowers within a narrow early-morning window.

This module employs smartphone-based image processing and deep learning techniques to detect pumpkin flowers and classify their gender under real field conditions. Images captured by the mobile application are processed using a convolutional neural network–based object detection model. The model learns discriminative morphological features such as ovary presence at the base of the flower, petal structure, flower size, and shape to distinguish between male and female flowers. Bounding box localization is applied to identify the spatial position of each detected flower, enabling precise targeting during artificial pollination.

To improve robustness, the training dataset includes images captured under varying lighting conditions, flower orientations, and growth stages. Data augmentation techniques are applied to address class imbalance and environmental variability. During inference, Non-Maximum Suppression is used to eliminate duplicate detections and retain high-confidence predictions. Only detections exceeding a predefined confidence threshold are forwarded to the pollination readiness module.

The integration of accurate flower and gender detection significantly reduces reliance on manual inspection and farmer experience. By providing reliable, real-time identification of receptive female flowers, this module enhances pollination timing accuracy, improves fruit set consistency, and supports efficient raised-bed ground artificial pollination in pumpkin farming.

Algorithm 1: Flower and Gender Detection Using Smartphone Images

Input: Smartphone-captured pumpkin flower image I
Output: Detected flower set

$$\mathcal{B} = \{(x, y, w, h, c, p)\} \quad (12)$$

Where (x, y, w, h) denote bounding box coordinates, c is the flower class (male or female), and p is the confidence score.

Steps:

1. Capture image I using the smartphone camera during the early-morning pollination window.
2. Preprocess I by resizing, normalization, and noise reduction.
3. Forward the preprocessed image through the trained CNN - YOLO detection mode
4. Divide the image into an $S \times S$ grid and predict bounding boxes and class probabilities for each grid cell.
5. For each bounding box prediction, compute the final confidence score using:
$$p^{final} = p^{obj} \times p^{class} \quad (13)$$
6. Apply Non-Maximum Suppression (NMS) to remove overlapping detections based on the Intersection over Union (IoU) threshold.
7. Retain detections satisfying $p^{final} \geq \tau$
8. Classify each retained detection as a male or female flower based on the predicted class label.

9. Output the bounding box coordinates, flower gender, and confidence score.
10. Forward high-confidence female flower detections to the pollination readiness module.

V. Discussion

The experimental results demonstrate that the proposed Smart Pollination Assistant can effectively support pumpkin pollination under real field conditions. High flower and gender classification accuracy confirms the suitability of smartphone-based deep learning models for real-time agricultural decision support [1][27][19]. The integration of environmental sensing enabled accurate identification of pollination-ready periods, leading to improved fruit set rates and reduced reliance on manual labor.

Sensor performance results indicate that low-cost IoT devices are sufficient for capturing the microclimatic conditions required for pollination readiness determination. The system's ability to terminate pollination recommendations after the viable window further improved operational efficiency by preventing unnecessary labor.

Compared to traditional hand pollination practices and existing pollination assistance approaches, the proposed system offers a cost-effective and scalable solution tailored to small and medium-scale farmers [10][20][22]. The use of active learning allows the model to adapt to local field conditions, addressing one of the major limitations of static smart agriculture systems.

A. Limitations

Despite promising results, several limitations were identified. The image processing model performance may degrade under extreme lighting variations or severe occlusion of flowers. Disease detection capabilities are currently limited to visible symptoms and do not account for early-stage or non-visual plant stress factors. Additionally, the system relies on smartphone availability and basic digital literacy, which may vary among farmers.

The experimental evaluation was conducted on a limited number of cultivation plots, and long-term seasonal performance variations were not fully analysed.

B. Future Work

Despite the promising results, several limitations were identified. The image processing model may experience reduced accuracy under extreme lighting variations, heavy occlusion, or severe weather conditions. Disease detection is currently limited to visible symptoms and does not account for early-stage or non-visual plant stress factors. Additionally, the experimental evaluation was conducted on a limited number of cultivation plots and over a single growing period, which may not fully capture seasonal variability. The system also depends on smartphone availability and basic digital literacy, which could affect adoption in certain farming communities.

VI. Conclusion

This paper presented a Smart Pollination Assistant for pumpkin farming that integrates Internet of Things–based environmental sensing, smartphone-based image processing, and adaptive machine learning. The proposed system enables automated flower and gender detection, real-time pollination readiness assessment, and timely decision support under field conditions. [18] [28] [29]

Experimental results demonstrated high flower classification accuracy, reliable sensor performance, improved fruit set rates, and reduced labor dependency. The integration of active learning further enhanced system adaptability to local farming conditions. Overall, the proposed approach provides a low-cost, scalable, and practical solution for pollination management in small and medium-scale pumpkin farming. The system contributes to sustainable smart agriculture by improving yield consistency and supporting data-driven agricultural decision making.

VII. References

[1] *"Automated Image Recognition of Staminate and Pistillate Flowers in Cucurbit Crops for Precision Pollination and Growth Management"*, Int. J. Environ. Sci., vol. 11, no. 9s, pp. 185–191, Jun. 2025, https://doi.org/10.64252/spb7fs28

[2] L.A.Garibaldi et al., "*Global dependence of agriculture on pollinators,"* Proc. Natl. Acad. Sci., vol. 108, no. 14, pp. 5909–5914, 2011. https://www.fao.org/pollination/about/en

[3] S. G. Potts et al., *"Safeguarding pollinators and their values to human well-being,"* Nature, vol.540, pp. 220–229, 2016. https://pubmed.ncbi.nlm.nih.gov/27894123/

[4] Aizen, M. A., Garibaldi, L. A., Cunningham, S. A., & Klein, A. M. (2008). "*Long-term global trends in crop yield and production reveal no current pollination shortage but increasing pollinator dependency"*. Current biology : CB, 18(20), 1572–1575. https://doi.org/10.1016/j.cub.2008.08.066

[5] Zhang, Q., Zhang, Z., Manzoor, S., Li, T., Igathinathane, C., Li, W., Zhang, M., Mhamed, M., Javidan, S., & Abdelhamid, M. (2025). A comprehensive review of autonomous flower pollination techniques: Progress, challenges, and future directions. *Computers and Electronics in Agriculture*, *237*, 110577. https://doi.org/10.1016/j.compag.2025.110577

[6] Kearns, C. A., Inouye, D. W. & Waser, N. 1998 "*Endangered mutualisms: the conservation of plant–pollinatorinteractions*". Annu. Rev. Ecol. Syst. 29, 83–112. doi:10.1146/annurev.ecolsys.29.1.83

[7] W. A. Inoka , P. Karunaratne & Jayanthi P. Edirisinghe (2006)*" Bee diversity in a semi-agricultural field - Sri Lanka"*, Biodiversity, 6:4, 17-20, http://dx.doi.org/10.1080/14888386.2005.9712781

[8] Shahab, H., Naeem, M., Iqbal, M., Aqeel, M., & Ullah, S. S. (2025). *"IoT-driven smart agricultural technology for real-time soil and crop optimization"*. Smart Agricultural Technology, 10, 100847. https://doi.org/10.1016/j.atech.2025.100847

[9] Broussard, M. A., Coates, M., & Martinsen, P. (2023). "*Artificial pollination technologies"*: A review. Agronomy, 13(5), 1351. https://www.mdpi.com/2073-4395/13/5/1351

[10] Sáez, A., Negri, P., Viel, M., & Aizen, M. A. (2019). *"Pollination efficiency of artificial and bee pollination practices in kiwifruit"*. Scientia Horticulturae, 246, 1017-1021. https://www.sciencedirect.com/science/article/pii/S0304423818308525

[11] Shuler, R. E., Roulston, T. A. H., & Farris, G. E. (2005). *"Farming practices influence wild pollinator populations on squash and pumpkin"*. Journal of economic entomology, 98(3), 790-795. https://academic.oup.com/jee/article-abstract/98/3/790/874491

[12] Stoner, K. A. (2020). *"Pollination is sufficient, even with low bee diversity, in pumpkin and winter squash fields"*. Agronomy, 10(8), 1141. https://www.mdpi.com/2073-4395/10/8/1141

[13] Petersen, J. D., Reiners, S., & Nault, B. A. (2013). *"Pollination services provided by bees in pumpkin fields supplemented with either Apis mellifera or Bombus impatiens or not supplemented"*. PLoS One, 8(7), e69819. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0069819

[14] Gavilánez-Slone, J. M. (2000). *"Pollination and pollinators of pumpkin and squash (Cucurbita maxima Duchesne) grown for seed production in the Willamette Valley of Western Oregon"*.

https://ir.library.oregonstate.edu/concern/graduate_thesis_or_dissertations/db78tf34k

[15] Wang, S., Yang, Y., Yin, H., Zhao, J., Wang, T., Yang, X., Ren, J., & Yin, C. (2025). *"Towards Digital Transformation of Agriculture for Sustainable Development in China*": Experience and Lessons Learned. Sustainability, 17(8). https://doi.org/10.3390/su17083756

[16] N. Suma, S. R. Samson, S. Saranya, G. Shanmugapriya, and R. Subhashri*,"International Journal on Recent and Innovation Trends in Computing andCommunication IOT Based Smart Agriculture Monitoring System,"* 2017, http://www.ijritcc.org

[17] Mohiuddin, M., Islam, M. S., & Shanjida, S. (2024). *Internet of Things (IoT)-Based Smart Agriculture Irrigation and Monitoring System Using Ubidots Server*. Engineering Proceedings, 82(1), 99. https://doi.org/10.3390/ecsa-11-20528

[18] Prakash, C., Singh, L. P., Gupta, A., & Lohan, S. K. (2023). Advancements in smart farming: *A comprehensive review of IoT, wireless communication, sensors, and hardware for agricultural automation*. Sensors and Actuators A: Physical, 362, 114605. https://doi.org/10.1016/j.sna.2023.114605

[19] Shi, W., Zhu, H., Lu, M., Li, J., & Han, Z. (2025). *A pumpkin seed vitality detection model based on deep spectral features.* Computers and Electronics in Agriculture, 236, 110457. https://doi.org/10.1016/j.compag.2025.110457

[20] He, L., Liu, X., Ding, Y., Jing, X., Dang, H., Murengami, B. G., Janneh, L. L., Li, R., Fountas, S., Arnal Barbedo, J. G., & Fu, L. (2025). *Advancements in artificial pollination of crops: From manual to autonomous.* Computers and Electronics in Agriculture, 231, 110067. https://doi.org/10.1016/j.compag.2025.110067

[21] Sinu, P. A., Pooja, A., & Aneha, K. (2019). *Overhead sprinkler irrigation affects pollinators and pollination in pumpkin (Cucurbita maxima). Scientia Horticulturae*, 258, 108803. https://doi.org/10.1016/j.scienta.2019.108803

[22] Cao, B., Xiang, Q., Jiao, H., Huang, X., Li, H., Yu, J., Guo, L., & Jiang, S. (2025). *Mechanized hand duster pollination maintains key fruit traits and mogroside V biosynthesis in Siraitia grosvenorii. Scientia Horticulturae*, 355, 114576. https://doi.org/10.1016/j.scienta.2025.114576

[23] Vogel, C., Mkandawire, T., Mkandawire, M., Küstner, G., Dakishoni, L., Bezner Kerr, R., Iverson, A., Poveda, K., & Steffan-Dewenter, I. (2023). *The importance of shrubland and local agroecological practices for pumpkin production in sub-Saharan smallholdings*. Basic and Applied Ecology, 73, 51-61. https://doi.org/10.1016/j.baae.2023.10.004

[24] Ezin, V., Gbemenou, U. H., & Ahanchede, A. (2022). *Characterization of cultivated pumpkin (Cucurbita moschata Duchesne) landraces for genotypic variance, heritability and agro-morphological traits*. *Saudi Journal of Biological Sciences*, *29*(5), 3661-3674. https://doi.org/10.1016/j.sjbs.2022.02.057

[25] Koocheki, A., Moghaddam, P. R., & Seyyedi, S. M. (2019). Saffron-pumpkin/watermelon: *A clean and sustainable strategy for increasing economic land equivalent ratio under limited irrigation*. *Journal of Cleaner Production*, *208*, 1327-1338. https://doi.org/10.1016/j.jclepro.2018.10.209

[26] Santos, A., Payne, R., Branco, M., & Franco, J. C. (2026). *Agricultural landscapes dominated by non-pollinator-dependent crops may play a role on the conservation of wild bees in riparian areas. Agriculture, Ecosystems & Environment*, *399*, 110213. https://doi.org/10.1016/j.agee.2025.110213

[27] Panchal, A. V., Patel, S. C., Bagyalakshmi, K., Kumar, P., Khan, I. R., & Soni, M. (2022). *Image-based Plant Diseases Detection using Deep Learning*. *Materials Today: Proceedings*, *80*, 3500-3506. https://doi.org/10.1016/j.matpr.2021.07.281

[28] Morchid, A., Muhammad Alblushi, I. G., Khalid, H. M., El Alami, R., Sitaramanan, S. R., & Muyeen, S. (2024). *High-technology agriculture system to enhance food security: A concept of smart irrigation system using Internet of Things and cloud computing*. *Journal of the Saudi Society of Agricultural Sciences*. https://doi.org/10.1016/j.jssas.2024.02.001

[29] Huda, S. S., Akhtar, A., Ahmed, E., Samiul Hoq, K. M., & Islam, M. N. (2026). *Artificial intelligence in agriculture across south Asia: Technology adoption, improvements, and sustainability outcomes. Sustainable Futures*, *11*, 101620. https://doi.org/10.1016/j.sftr.2025.101620

[30] Zarboubi, M., Bellout, A., Chabaa, S., & Dliou, A. (2026). Enhancing integrated pest management with IoT and YOLO-Evo: A smart, low-cost monitoring system for sustainable apple farming. *Results in Engineering*, *29*, 108850. *https://doi.org/10.1016/j.rineng.2025.108850*

[31] Morchid, A., Ismail, A., Khalid, H. M., Qjidaa, H., & Alami, R. E. (2025). Blockchain and IoT technologies in smart farming to enhance the efficiency of the agri-food supply chain: A review of applications, benefits, and challenges. *Internet of Things*, *33*, 101733. https://doi.org/10.1016/j.iot.2025.101733

[32] Krumbe, F., Melder, S., & Feuerbacher, A. (2025). The vital role of pollination services in seed production: A global review. *Agriculture, Ecosystems & Environment*, *393*, 109745. https://doi.org/10.1016/j.agee.2025.109745

[33] Blili, Y., Abou Nader, E., Pavez, I., Prosperi, P., Harbouze, R., Kyrgiakos, L. S., Kleisiari, C., Vasileiou, M., Angelopoulos, V., Vlontzos, G., & Kleftodimos, G. (2025). Assessing the economic impact of insect pollination on the agricultural sector: A department-level case study in France. *Environmental and Sustainability Indicators*, *28*, 100944. https://doi.org/10.1016/j.indic.2025.100944